



Intelligent Transportation Systems and Traffic Operations

Introduction

Instructor

Heeseung Shon, Ph.D.

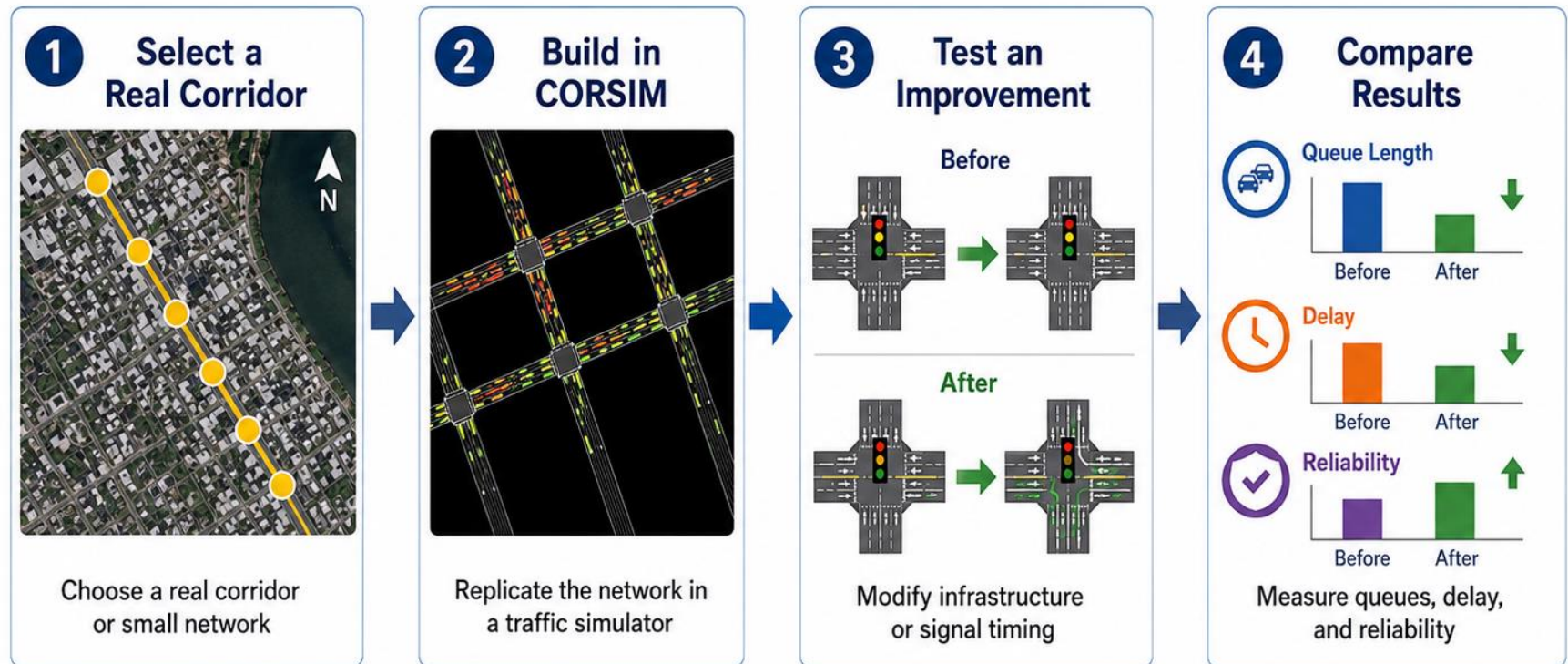
Course Objectives

□ **This course aims to develop students' knowledge and skills in:**

- ❖ **Foundations of transportation engineering:** construct and read time-space diagrams; analyze queues, congestion, and capacity using queueing and traffic-flow theory; design and evaluate traffic signal timing.
- ❖ **Network modeling and trip assignment:** predict travel demand with the four-step model; perform and interpret user-equilibrium and system-optimal assignments; explain network paradoxes.
- ❖ **Smart mobility and emerging technologies:** Explain next-generation transportation; explore AI methods for predicting traffic speed and flow; assess the sustainability of transportation modes.

Mini Project

- ❖ Select a real corridor or small network; replicate it in a traffic simulator; implement an infrastructure change or a signal plan to improve traffic conditions; document changes in queues, delay, and reliability.



Example tool: CORSIM

Textbook and References

- ❑ **Lecture Note**

- ❑ **Principles of Highway Engineering and Traffic Analysis
(Mannering and Washburn), Chapters 5-8.**

Class Contents

What is Transportation Engineering?

Time-Space Diagram

Traffic Properties & Measurement

Queuing Theory & Analysis

Traffic Flow Theory

Signal System (Cyclic System)

4-Step Model

Trip Assignment

Network Paradoxes

Cooperative-ITS (C-ITS)

Future Transportation Modes



Intelligent Transportation Systems and Traffic Operations

What is Transportation Engineering?

Instructor

Heeseung Shon, Ph.D.

What is Transportation Engineering?

Overview

□ Definition

- All the activities involved in carrying, moving or conveying either **people** or **goods**

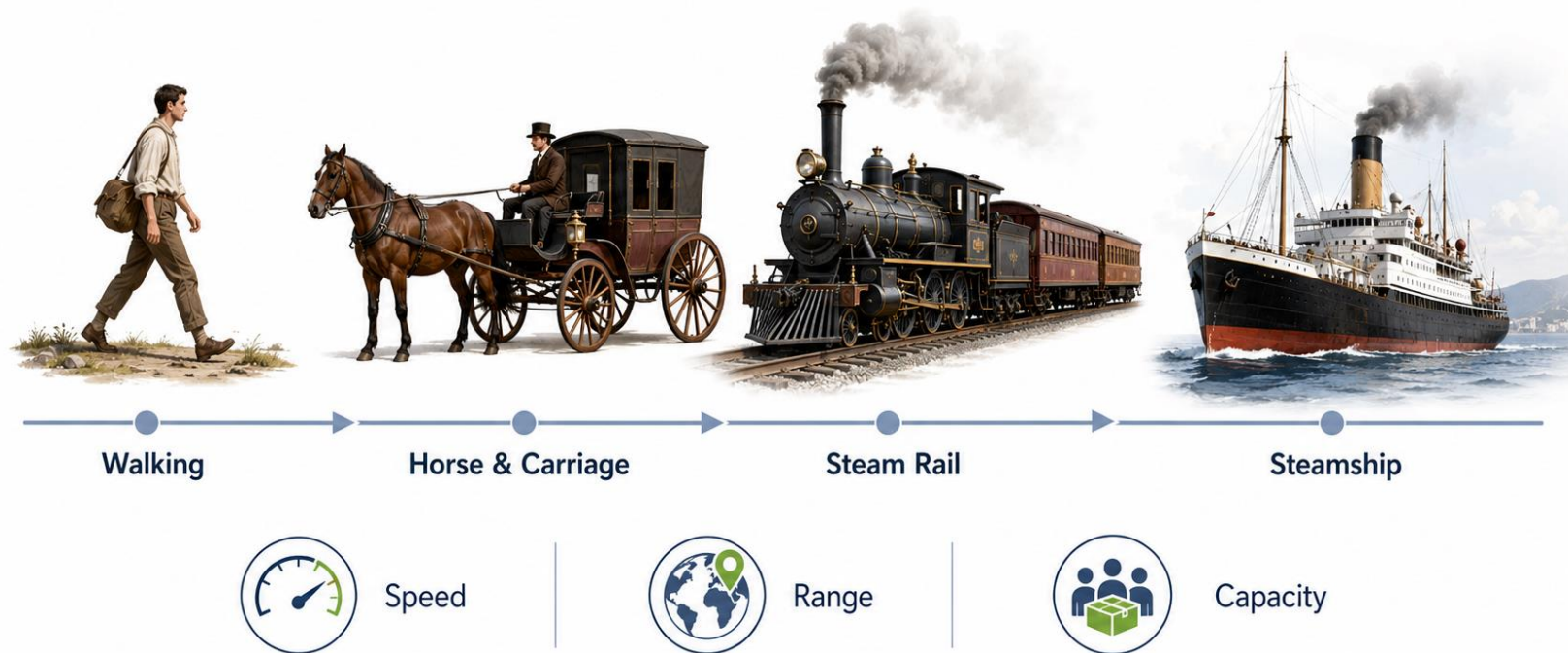


What is Transportation Engineering?

History of Transportation

❑ From human power to mechanized transportation

- Transportation gradually reduced the constraints of **distance** and **time**.



What is Transportation Engineering?

History of Transportation

❑ The Automobile: Expanding Access to Opportunities

- More destinations could be reached within the same amount of time.
- Access to employment, education, markets, and healthcare expanded.

Horse-drawn travel
30-minute reachable area



Automobile travel
30-minute reachable area



The value of transportation is not movement itself, but **access** to **opportunities**.

What is Transportation Engineering?

History of Transportation

❑ Transportation Challenges

1 TRAFFIC CONGESTION AND DELAY



\$85.8 BILLION
ANNUAL COST OF TRAFFIC
CONGESTION IN THE U.S.



49 HOURS
lost per typical U.S. driver
in 2025



112 HOURS
lost per driver in Chicago
in 2025

Source: INRIX 2025 Global Traffic Scorecard (U.S.)

2 CRASHES AND SAFETY RISKS



39,345
PEOPLE KILLED IN TRAFFIC CRASHES
IN THE U.S. IN 2023



\$340 BILLION
ESTIMATED SOCIETAL COST
OF MOTOR VEHICLE CRASHES
IN 2023

Sources: NHTSA (2023), NCSA (2023)

3 ENERGY CONSUMPTION AND EMISSIONS



1.6 BILLION METRIC TONS
CO₂ EMISSIONS FROM
TRANSPORTATION IN THE U.S. (2022)
~27% OF TOTAL U.S. CO₂ EMISSIONS



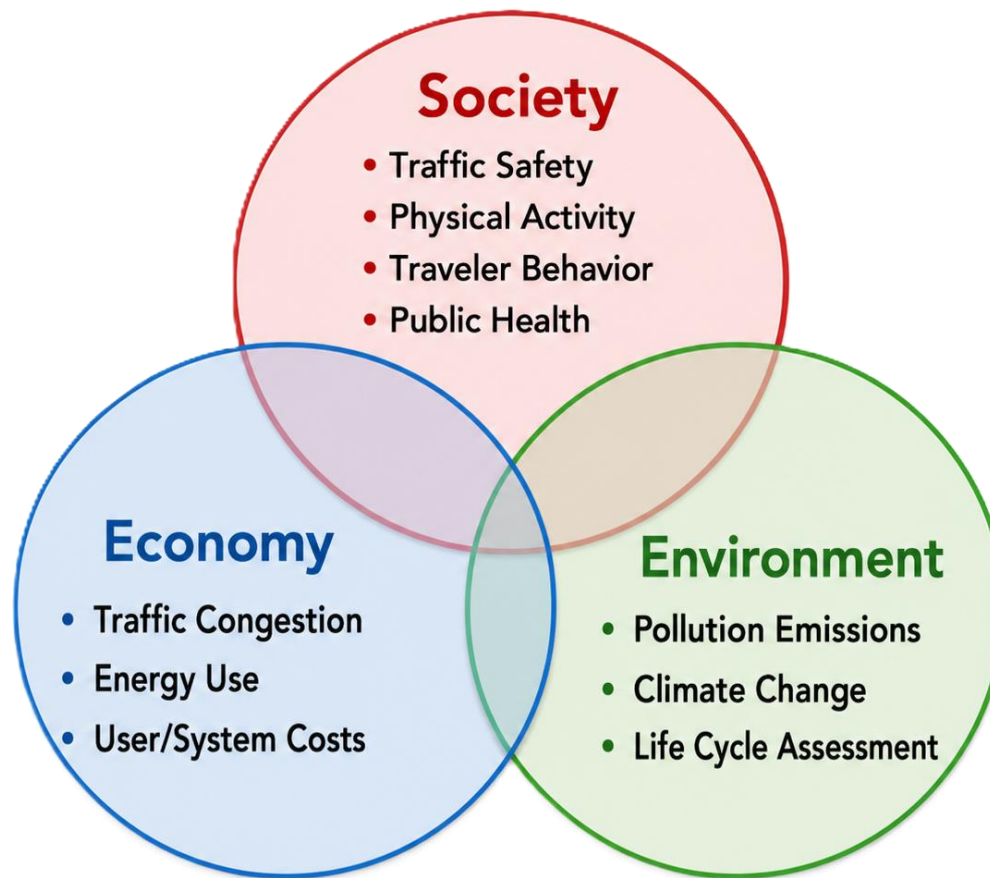
~19.7 BILLION GALLONS
GASOLINE CONSUMED IN THE U.S. (2022)

Sources: EPA (2022), EIA (2022)

What is Transportation Engineering?

History of Transportation

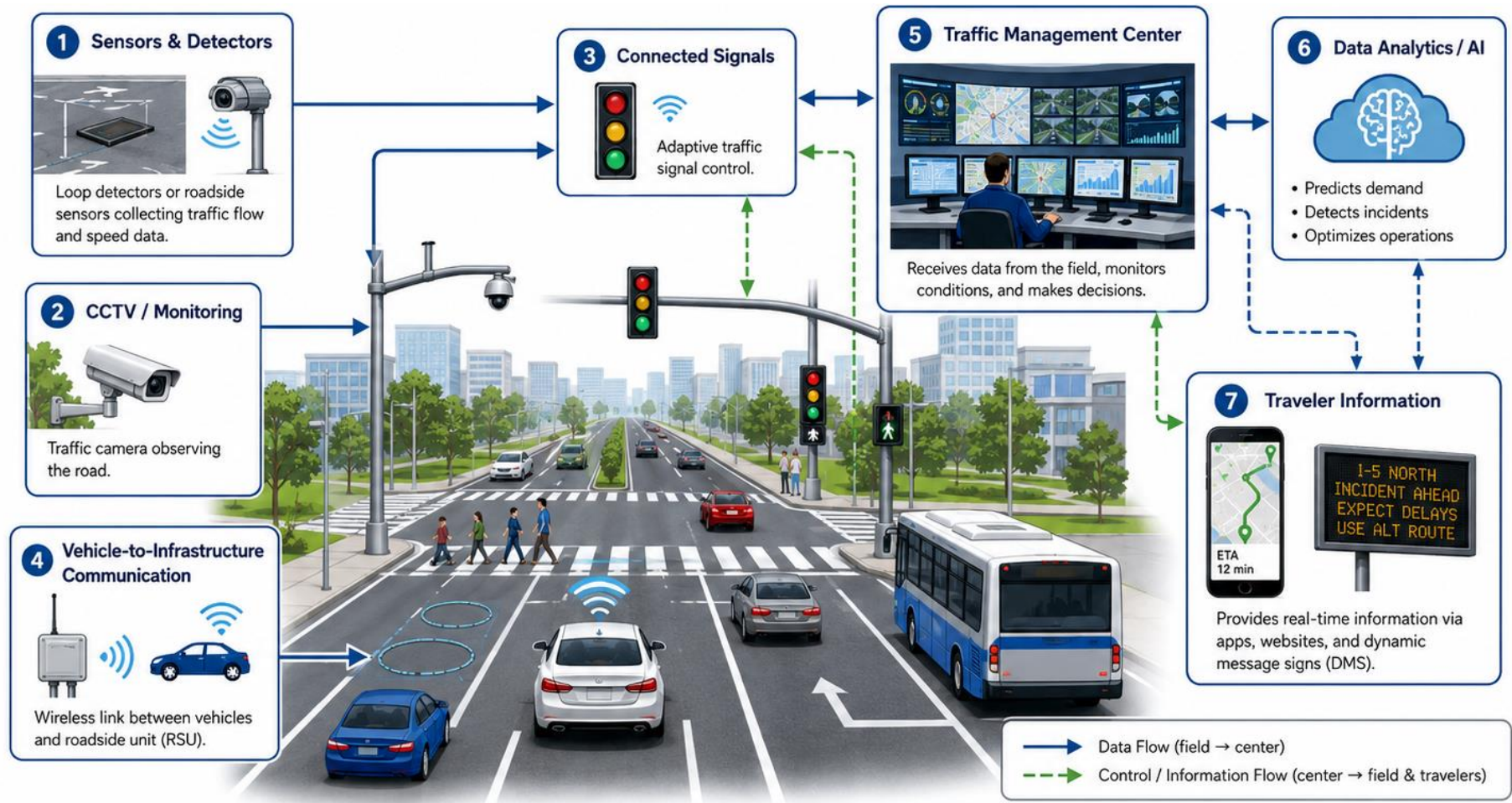
❑ Sustainable Transportation



What is Transportation Engineering?

History of Transportation

□ Intelligent Transportation Systems (ITS)



What is Transportation Engineering?

History of Transportation

❑ Five primary functional area of ITS

- Advanced Traffic Management Systems (ATMS)
- Advanced Traveler Information Systems (ATIS)
- Commercial Vehicle Operations (CVO)
- Advanced Public Transportation Systems (APTS)
- Advanced Vehicle Control Systems (AVCS)

What is Transportation Engineering?

Future Transportation

❑ Cooperative-ITS (C-ITS)



What is Transportation Engineering?

Future Transportation

❑ Advanced Air Mobility Systems

- Electric Vertical Take-Off and Landing (eVTOL)



eVTOL



Vertiport

What is Transportation Engineering?

Future Transportation

❑ Advanced Air Mobility Systems



Port of Doha → Katara Cultural Village
30 min by car → 8 min by eVTOL
Human-carrying pilotless trial flight, 2025